INTERACTIVE SINHALA TEXT READER FOR THE VISUALLY IMPAIRED USING OCR, NLP, AI.
R26-DS-002

Project Proposal Report

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Sri Lanka Institute of Information Technology
Sri Lanka

March 2026

ADVANCE DOCUMENT RETRIEVAL AND INFORMATION GENERATION
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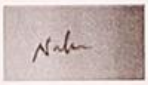
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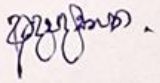
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

For the estimated 500,000 visually disabled individuals in Sri Lanka, independent access to printed Sinhala newspapers remains largely out of reach. Existing commercial assistive devices such as OrCam MyEye and Envision Glasses carry prices exceeding $1,800 and provide no Sinhala language support, while freely available applications like Google Lookout require physical dexterity and deliver content without any adjustment for the reader's comprehension level. Academic research in Sinhala OCR has advanced recognition accuracy meaningfully, yet every published system treats corrected text as its final output none addresses what should happen downstream in terms of retrieval, contextual understanding, or adaptive delivery.

Three specific gaps drive this research. No computational framework for measuring Sinhala text complexity exists anywhere in the published literature. No retrieval architecture has been designed to handle the character-level noise that Sinhala OCR characteristically introduces. And no Sinhala knowledge base has been built to update itself without human intervention.

This research proposes an integrated Sinhala document retrieval and generation system addressing all three gaps within a single deployable pipeline. A headline-anchored two-stage retrieval mechanism using LaBSE embeddings and body-text cosine reranking handles OCR-degraded input. A three-tier self-evolving ChromaDB knowledge base populated from Lankadeepa and Mawbima with automatic reliability-scored content promotion maintains current Sinhala news content without manual curation. The first Sinhala Text Complexity Framework, trained on a 150-paragraph native-speaker annotated corpus using logistic regression over four linguistically motivated features, enables complexity-conditioned content adaptation. A LangChain and Gemma 2B RAG pipeline delivers simplified or elaborate Sinhala responses based on user intent signals from Component 4, operating fully offline on standard consumer hardware.

Expected outcomes include a measurable retrieval precision improvement over single-stage baselines, a validated Sinhala complexity classifier with reported accuracy and F1, and a user preference alignment rate confirmed through a nine-person native speaker study. The system deploys hardware costing under $50, advancing SDG 4, SDG 10, and SDG 11 by providing equitable access to daily print information for visually impaired Sinhala readers.

Keywords: Sinhala NLP, Retrieval-Augmented Generation, Assistive Technology, Text Complexity Classification, Visually Impaired

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
LLM	Large Language Model
NLP	Natural Language Processing
JSON	JavaScript Object Notation
API	Application Programming Interface
ChromaDB	Chroma Vector Database
CSV	Comma Separated Values
HAL	Hal Kirima (Sinhala Vowel Suppression Sign)
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
KB	Knowledge Base
LaBSE	Language-Agnostic BERT Sentence Embeddings
NFC	Unicode Normalization Form C
OCR	Optical Character Recognition
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
sinling	Sinhala Language Processing Library
URL	Uniform Resource Locator
TTR	Type-Token Ratio

1. INTRODUCTION

1.1 Overall Background of Research

Across most of the world, picking up a newspaper and reading it independently is so unremarkable an act that it rarely registers as a privilege. Yet for the approximately 2.2 billion people who live with some degree of visual impairment or blindness globally, that same act depends almost entirely on whether someone else is willing and available to help [1]. In Sri Lanka, where government census data places the visually disabled population at over 500,000 individuals, this dependence shapes daily life in ways that extend well beyond the inconvenience of missed headlines. It touches employment, civic participation, health literacy, and the basic sense of keeping pace with one's own community.

Sinhala, spoken natively by roughly 74 percent of Sri Lankans, holds an obvious centrality to how most people in the country receive information [3]. Newspapers such as Lankadeepa and Mawbima form part of the daily rhythm of millions of households, carrying reporting on politics, local affairs, health, and education that many readers treat as their primary window onto the wider world [4]. A visually impaired person who cannot access these publications independently is, in a meaningful sense, cut off from that window reliant on the goodwill and availability of others rather than on tools built to serve them directly. The United Nations has framed access to education and information as foundational to sustainable development, and the barriers described here sit squarely in tension with SDG 4, SDG 10, and SDG 11, each of which speaks to inclusion, equity, and accessible public participation [5].

This research grows out of that tension. Component 3 of a four-part assistive reading pipeline addresses the retrieval, comprehension, and adaptive delivery of Sinhala newspaper content for visually impaired users. Positioned after optical character recognition and error correction stages, it receives cleaned Sinhala text and generates responses calibrated to the individual reader's comprehension level. The work draws on natural language processing, information retrieval, and machine learning within the Autonomous Intelligent Machines and Systems research cluster and the Data Science specialization fields well suited to a problem that is both technically demanding and socially consequential.

1.1.1 Relevant Studies and Techniques

The literature bearing on this research falls into three areas that have, until recently, developed with relatively little cross-pollination: commercial assistive hardware, Sinhala-specific optical character recognition, and the broader field of retrieval-augmented language generation.

Wearable assistive devices represent the most visible contemporary response to the reading access problem. OrCam MyEye and Envision Glasses both demonstrate that compact glass-mounted cameras can be paired with character recognition and speech synthesis to give visually impaired

users some degree of reading independence in everyday environments [6], [7]. The engineering achievement these products represent is genuine. What they establish, just as usefully, is a design vocabulary capture, recognize, and say that subsequent systems have largely inherited, including the broader assistive pipeline within which this research sits. Where the present work departs from that lineage is in what happens between recognition and speech: the question of whether the recognized content can be retrieved, contextualized, and shaped to suit the person listening.

Sinhala OCR has attracted a modest but growing body of academic attention. The work of Anuradha, Liyanage, and Weerasinghe offered one of the first rigorous examinations of how document genre, scanning resolution, and algorithm selection each influence recognition outcomes for Sinhala text, with display-weight typography emerging as a particularly stubborn source of error [8]. Ambegoda's subsequent benchmarking study placed several current recognition frameworks Tesseract, Surya, and purpose-built LSTM models into direct comparison on Sinhala and Tamil material, finding that conjunct character clusters and the column-heavy layout conventions of Sri Lankan print journalism continued to trip up even the strongest evaluated systems [9]. Both studies contribute meaningfully to understanding where and why recognition fails. Neither extends its analysis to what a downstream system ought to do with imperfectly recognized text once it has been produced.

That downstream question has been approached from a different direction entirely by the natural language processing community. Lewis and colleagues proposed retrieval-augmented generation as a way of anchoring language model outputs to externally sourced factual material, demonstrating that this grounding produces measurably better performance on knowledge-intensive tasks than relying on model parameters alone [10]. Separately, Feng and colleagues introduced LaBSE a sentence embedding model trained across 109 languages simultaneously and showed that the resulting representations maintain semantic coherence even across language pairs with limited parallel data [11]. For a retrieval system operating on Sinhala text that an OCR process has already partially corrupted, the noise-tolerance implicit in meaning-level rather than character-level comparison is not a minor convenience. It is a foundational requirement, and LaBSE's multilingual training makes it the most practically viable embedding model available for this purpose.

1.1.2 State of Art

Mapping the current landscape of assistive reading technology for Sinhala speakers requires holding two realities in mind simultaneously. The first is that the broader assistive technology field has made genuine and impressive strides wearable devices, multilingual NLP libraries, and sophisticated language models that now exist that would have seemed implausible a decade ago. The second is that almost none of this progress has reached Sinhala in any form that a visually impaired Sri Lankan reader could practically use today.

At the commercial end of the spectrum, OrCam MyEye and Envision Glasses stand as the most capable currently available wearable reading devices, combining text detection, optical character recognition, and speech output in hardware small enough to mount on an ordinary pair of glasses [6], [7]. Neither product supports Sinhala. Their pricing structures, which regularly place the hardware above $1,800 per unit, add a second layer of inaccessibility that is entirely independent of the language question a Sri Lankan household at median income would need to commit several months of earnings to acquire either device. Google Lookout approaches the problem from a different angle, offering document scanning through a smartphone camera at no cost to the user [12]. Its practical ceiling is visible almost immediately, however: the application requires a steady handheld phone, produces linear audio output with no interactive features, and makes no attempt to adjust what it reads based on who is listening.

Sinhala NLP tooling tells a similar partial story. The sinling library, maintained at the University of Moratuwa, provides the most complete publicly available infrastructure for Sinhala text processing tokenization, part-of-speech analysis, and basic morphological handling are all within its scope [13]. Complexity measurement is not. Separate work on Sinhala embedding architectures has shown that two-tiered models combining word-level and sentence-level representations outperform single-tier approaches on Sinhala sentiment classification tasks, confirming that the language yields to principled computational analysis when appropriate tools are applied [14]. What that work does not yield is any resource for measuring how demanding a passage of Sinhala text is to read a capability that the present research treats as foundational rather than supplementary.

The readability measurement field in English offers the clearest indication of what is missing. Instruments like the Flesch-Kincaid Grade Level formula and the SMOG index have been in practical use since the mid-twentieth century, applied across educational publishing, health communication, and legal plain-language initiatives on the basis of a well-established finding: that surface features of text sentence length above all, followed by vocabulary characteristics predict reader-perceived difficulty with sufficient reliability to be useful [15]. The theoretical reasoning behind these instruments' transfers readily to other languages. The empirical work of building, calibrating, and validating equivalent instruments for Sinhala has simply not been done. A system that claims to adjust content difficulty for different Sinhala readers without any measurement of what difficulty currently means in that language is making an assertion it cannot support. Closing that gap is the central contribution this research proposes.

1.2 Background & Literature Survey

1.2.1 Overview of the Domain

Technological progress rarely distributes its benefits evenly. Over the past two decades, advances

in computer vision, natural language processing, and speech synthesis have transformed what assistive tools can do yet the population standing to gain most from these advances have often been among the last to see them arrive in usable form. For speakers of low-resource languages, the delay is structural rather than accidental. Building language support into a product requires investment, and investment follows markets. Sinhala, despite being spoken by most of a nation of over 22 million people, has not attracted the kind of sustained commercial attention that would bring it within the scope of most assistive technology development programmers.

Three research areas converge in the work described in this proposal. Optical character recognition of printed Sinhala documents has been studied with increasing rigor, with investigators identifying conjunct character formation and the stylized typefaces typical of newspaper headlines as sources of recognition error that remain problematic even for contemporary deep learning pipelines [8], [9]. Separately, the introduction of retrieval-augmented generation offered a practical answer to a question that had long constrained language model deployment how to ground model outputs in verified factual content without retraining from scratch for every domain [10]. And running through decades of English-language educational research, the study of text readability has produced validated instruments for predicting how difficult a passage will be for a given reader, instruments whose theoretical basis extends to other languages even though the empirical work of building them for Sinhala has never been undertaken [15].

These three threads are not parallel interests. They address consecutive stages of what a functional assistive reading system must do. Recognition recovers the content. Retrieval locates and contextualizes it. Complexity measurement determines whether the recovered, contextualized content is usable by the person it is meant to serve. A system that handles all three stages in sequence and handles them in Sinhala would occupy genuinely new ground in the assistive technology landscape. That is the practical ambition driving this research.

1.2.2 Overall Background of the Problem

Two kinds of difficulty sit at the heart of this research, and neither can be properly understood without the other.

Sinhala is, from a computational linguistics standpoint, an awkward language to work with. Its writing system is alpha syllabic each character encodes a consonant paired with a vowel, rather than a single sound and the mechanism by which consonants combine into conjunct forms through the HAL sign creates visual structures that standard optical character recognition error models were not designed to handle [8]. The result is that even well-performing recognition systems produce error patterns in Sinhala that resist straightforward correction. Further along the processing chain, the scarcity of large, clean, verified Sinhala text collections limits what can be done with embedding models, generative systems, and any machine learning component that depends on labelled training data [14]. The sinling library, developed at the University of Moratuwa, offers the most complete publicly available set of Sinhala processing tools and handles tokenization and morphological analysis adequately [13]. It does not, however, provide any mechanism for assessing how complex a passage of Sinhala text is or for connecting a language

model's outputs to an external factual knowledge source capabilities that this research treats as central rather than peripheral.

The people most affected by these technical absences are those who had the least access to print information to begin with. Sri Lanka's visually disabled population numbers over 500,000 by government estimates, and the practical reality for most of these individuals is that reading a newspaper means finding someone willing to read it aloud for them [2]. This is not a trivial constraint. Publications like Lankadeepa carry the kind of daily civic information health advisories, educational announcements, political developments, local reporting that most readers treat as essential rather than supplementary [4]. Being excluded from that information flow consistently, day after day, is an exclusion from public life in a meaningful sense. Commercial hardware exists that could, in principle, reduce this exclusion. OrCam MyEye and Envision Glasses both offer document reading functionality, but neither supports Sinhala and both carry retail prices that make them financially inaccessible to the households where need is greatest [6], [7]. Google Lookout costs nothing, which removes one barrier while leaving others intact it requires users to hold a phone steady throughout the scanning process, a physical demand that is unreliable for many visually impaired users, and it reads whatever text it finds without any consideration of whether the reader can follow it [12].

What makes this research necessary is not one of these problems in isolation but the way they reinforce each other. The technical gaps mean that a genuinely capable Sinhala assistive reading system does not yet exist. The social conditions mean that the absence of such a system carries real costs for many people who have no adequate alternative.

1.2.3 Significance of the Research

The proposed component addresses five critical dimensions absent from existing systems:

1. Fills a gap that no prior Sinhala NLP work has addressed

Every language that has developed mature assistive reading technology has, somewhere along the way, produced a method for measuring how difficult its text is to read. Sinhala has not. No published computational tool for classifying Sinhala text complexity exists in the literature [13], [15]. Without one, any system claiming to adapt content for different readers is operating on assumption rather than measurement. This research builds and validates that missing tool for the first time.

2. Generates five concrete, independently verifiable results

Rather than proposing a system and asserting its value, this research produces five measurable outputs that can be checked by any reviewer: a classifier accuracy and F1 comparison table, a retrieval precision benchmark across thirty-five OCR-noisy test articles, a knowledge base precision curve across one hundred simulated sessions, a user preference alignment rate from a nine-person native speaker study, and an end-to-end latency measurement on target hardware. Each stand as independent evidence.

3. Offers something that genuinely does not exist for this population

A visually impaired Sinhala speaker looking for an affordable, hands-free, interactive reading tool currently has no viable option [6], [7], [12]. This system runs on hardware costing under fifty dollars, operates without internet connectivity, and adjusts content complexity per user to a combination of capabilities that no existing product or published research system delivers together.

4. Moves Sinhala assistive research past the recognition boundary

Every Sinhala assistive study reviewed for this research treats correctly recognized text as its final output [8], [9]. This work begins where those studies end taking recognized text into retrieval, contextual generation, and reader-adapted delivery. That extension is not incremental. It represents a qualitatively different conception of what such a system should do.

5. Connects to commitments Sri Lanka has already made

The barriers this research addresses map directly onto three Sustainable Development Goals SDG 4, SDG 10, and SDG 11 each of which Sri Lanka is formally committed to advancing [5]. A system that gives visually impaired Sinhala readers independent access to daily print information contributes to all three in a way that is specific, demonstrable, and locally grounded.

6. Leaves behind a dataset the field can use

Building the complexity classifier required creating something that did not previously exist: a corpus of 150 Sinhala paragraphs rated for reading difficulty by native speakers, with inter-annotator agreement formally measured. That corpus has value beyond this research. Future work on Sinhala readability, language modelling, or educational NLP can draw on it directly a contribution that persists regardless of what becomes of the system itself.

1.3 Research Gap

The first and most consequential is the complete absence of any computational framework for measuring Sinhala text complexity. English readability research has produced validated instruments Flesch-Kincaid, SMOG, and related indices whose theoretical basis rests on the demonstrable relationship between surface textual features and reader-perceived difficulty [15]. No equivalent has been developed for Sinhala. The sinling library, the most complete Sinhala NLP toolkit currently available, provides tokenization and morphological analysis but contains no complexity measurement functionality [13]. This absence means that every existing Sinhala assistive system either ignores reading difficulty entirely or adjusts content based on principled criterion, a limitation that this research directly resolves by proposing and empirically validating the first feature-based Sinhala complexity classifier.

The second gap concerns retrieval over OCR-degraded input. Ambegoda's benchmarking work confirms that Sinhala OCR continues to produce character-level errors in conjunct sequences and

display typography [9]. Standard dense retrieval systems embed noisy text directly, producing shifted vectors that cause retrieval failure at rates that single-stage approaches cannot adequately suppress. No published work proposes a retrieval architecture specifically designed to handle this degradation in Sinhala document contexts.

The third gap is the absence of any Sinhala knowledge base capable of updating itself without human intervention. Lewis et al. established RAG as the dominant pattern for knowledge-grounded generation, but all reviewed implementations use static, manually curated corpora [10]. For a daily newspaper reading application, a static knowledge base becomes outdated immediately upon deployment a practical constraint that no existing Sinhala system has addressed.

Current approaches are insufficient not because they fail at what they attempt, but because what they attempt stops well short of what genuine assistive reading requires [8], [9]. The novelty of this research lies in addressing all three gaps within a single integrated pipeline validated through measurable experimental results.

1.4 System Comparison Table

Aspect	**Proposed System**	Anuradha et al. (2020) [8]	Ambegoda (2025) [9]	Lewis et al. (2020) [10]	Feng et al. (2022) [11]	Perera et al. (2021) [14]	sinling UoM [13]	OrCam MyEye [6]	Envision Glasses [7]	Google Lookout [12]
Sinhala Language Support	Yes	Yes	Sinhala	No	Partial	Yes	Yes	No	No	Sinhala
Handles OCR-degraded input	Yes	No	No	No	No	No	No	Yes		No
Document retrieval capability	Yes	NO	No	Yes-english	Yes	Yes	Yes	No		No
Text complexity measurement	Yes	No	No	No	No	No	Yes	Yes		No

Complexity classification	Yes	No	No	No	No	No	No	No		No
Annotated complexity dataset	Yes	No	No	No	No	Yes	Yes	Yes		No
RAG pipeline	Yes	NO	Yes - English	No	No	No	No	yes		No

Table 1. System Comparison Table

1.5 Research Problem

1.5.1 Primary Question

How can a Retrieval-Augmented Generation pipeline improve information

Visually impaired Sinhala speakers have no access to a system that can read a printed newspaper, retrieve contextually relevant information about its content, and deliver that content at a reading difficulty level appropriate for the individual listener without requiring sighted assistance, specialized expensive hardware, or an internet connection.

This absence is not explained by a single technical limitation. It reflects the compounding effect of several gaps occurring simultaneously. Commercial assistive devices capable of reading printed text aloud exist, but none support Sinhala and their cost places them beyond the reach of the demographic most affected [6], [7]. Sinhala OCR research has progressed to the point where printed text can be recognized with meaningful accuracy, but the recognition pipeline produces character-level errors particularly in conjunct sequences that corrupt downstream retrieval when standard embedding approaches are applied [8], [9]. Retrieval-augmented generation offers a principled framework for grounding language model responses in verified factual content, but no RAG system has been constructed for Sinhala in any published work [10]. And cutting across all of these problems sits a more fundamental one: no computational measure of Sinhala text complexity exists, meaning that even a system capable of retrieving and generating Sinhala content has no principled basis for adjusting what it delivers to match the person receiving it [13], [15].

The primary research problem, stated precisely, is therefore this: the field lacks an integrated Sinhala document retrieval and generation system that combines OCR-robust article matching, self-maintaining knowledge base management, and empirically grounded complexity-adaptive content delivery within a single deployable pipeline.

1.5.2 Specific Research Problems

The primary problem decomposes into three specific problems, each of which corresponds to a distinct gap in the existing literature and a distinct novelty contribution in the proposed system.

The absence of any Sinhala text complexity measurement tool

When OCR errors corrupt conjunct characters and diacritics in a scanned Sinhala document, the resulting noisy text shifts the embedding vector away from its clean counterpart in the semantic space [9]. Single-stage dense retrieval, which compares the full noisy query vector against stored article vectors, fails at rates that are difficult to suppress through threshold tuning alone. No published work addresses retrieval precision degradation caused specifically by Sinhala OCR noise, and no two-stage retrieval architecture designed to handle this failure mode has been reported for any Indic script language in an assistive reading context [8], [9], [11].

The absence of OCR-robust document retrieval for Sinhala

No computational framework for measuring Sinhala text reading difficulty exists in the published literature [13], [15]. English readability instruments such as Flesch-Kincaid rest on the empirically validated relationship between surface textual features and reader-perceived difficulty a relationship that has never been studied for Sinhala. Without such a framework, complexity-adaptive content delivery for Sinhala readers has no measurement foundation, making any claimed adaptation unverifiable and inconsistent across users and sessions.

The static nature of existing Sinhala knowledge bases

Every Sinhala NLP system identified in this review relies on a fixed, manually curated corpus that requires human intervention to update [10], [13].For a daily newspaper reading assistant, a static knowledge base becomes obsolete from the moment of deployment today's articles are simply absent. No self-updating mechanism for Sinhala content has been reported in any published system, leaving a practical gap between what research prototypes demonstrate in controlled conditions and what a real daily-use assistive system would require.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and empirically validate a complexity-adaptive Sinhala document retrieval and generation system that enables visually impaired users to access, query, and receive printed Sinhala newspaper content at a reading difficulty level matched to their individual comprehension needs operating fully offline on standard consumer hardware without requiring sighted assistance.

2.2 Specific Objectives

The main objective is pursued through six specific objectives, each addressing a distinct technical component of the proposed system and generating at least one independently measurable research result.

Sub Object 1 - To construct a verified Sinhala news article knowledge base through systematic web scraping and preprocessing

A corpus of no fewer than 2,000 Sinhala news articles will be collected from Lankadeepa and Mawbima using automated scraping, subjected to Unicode normalisation and sinling-based tokenisation, deduplicated, and stored as structured JSON objects in a locally deployed ChromaDB vector store. This corpus forms the Tier 1 foundation of the self-evolving knowledge base and constitutes the retrieval index against which all incoming scanned documents are matched.

Sub Object 2 - To develop and benchmark a headline-anchored two-stage retrieval pipeline for OCR-degraded Sinhala documents

A two-stage retrieval architecture will be implemented in which LaBSE headline embeddings perform coarse candidate selection followed by body-text cosine reranking over a three-sentence window. The pipeline will be evaluated against a single-stage baseline on a manually constructed ground-truth test set of thirty-five OCR-noisy Sinhala newspaper articles, with Top 1 and Top 3 precision reported for both approaches. The measured precision improvement constitutes the primary evidence for this objective.

Sub Object 3 - To design and implement a three-tier self-evolving knowledge base with automated content promotion

The knowledge base will operate across three tiers a permanent verified store, a session cache, and a web-scraped fallback governed by a reliability scoring mechanism through which Tier 3 content is automatically promoted to Tier 1 upon repeated successful retrieval without human intervention. Knowledge base quality improvement will be measured as a retrieval precision curve across one hundred simulated usage sessions, with precision recorded at five intervals.

Sub Object 4 - To construct and empirically validate the first Sinhala Text Complexity Framework

A manually annotated corpus of 150 Sinhala paragraphs, rated for reading difficulty across three levels by eight to ten native speakers, will be used to train a logistic regression complexity

classifier on four linguistically motivated features mean sentence length, type-token ratio, conjunct character ratio, and mean word length. Classifier performance will be reported as accuracy and F1 per class against a held-out test set of thirty samples, with a Random Forest comparison baseline included to justify model selection. A nine-participant native speaker user study will measure preference alignment rate across three demographic groups as external validation.

Sub Object 5 - To implement a signal-conditioned dynamic text adaptation module integrating the complexity framework with a RAG generation pipeline

A LangChain RetrievalQA chain connecting the ChromaDB knowledge base to a locally deployed Gemma 2B language model will be constructed and configured with level-specific Sinhala prompt templates. Upon receiving a simplification signal, the system will apply Level 1 constraints maximum ten-word sentences, active voice, simple vocabulary to the retrieved article. Upon receiving an elaboration signal, multi-article semantic retrieval will expand the context window to support background-enriched generation. Both modes will be tested against twenty manually prepared evaluation cases.

Sub Object 6 -To evaluate end-to-end system performance against defined thresholds on target hardware

The complete pipeline from Component 2 JSON input to complexity-adapted Sinhala text output will be executed across twenty test sessions on a Core i5 processor with 8 gigabytes of RAM. Mean and maximum end-to-end latency will be recorded against a target threshold of fifteen seconds per session. System performance will additionally be assessed through the five research results produced across SO2, SO3, SO4, and SO5.

3. METHODOLOGY

3.1 HIGH-LEVEL SYSTEM ARCHITECTURE

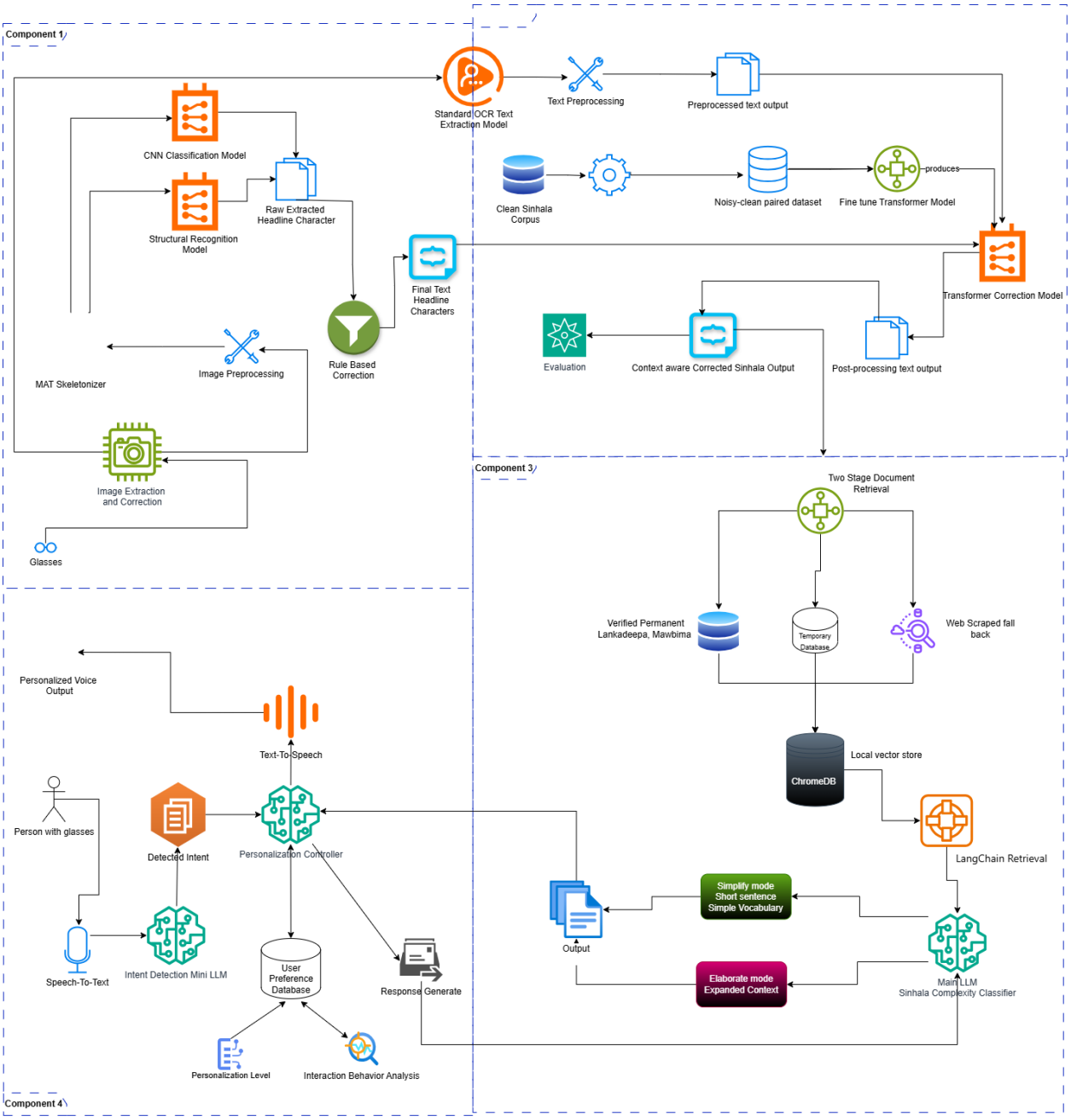


Figure 1.High Level System Architecture

3.2 System Overview

The proposed system operates as a four-component sequential pipeline in which each component performs a distinct transformation on the input before passing its output to the next stage. The physical entry point is a pair of glasses worn by the visually impaired user, equipped with a camera that captures the printed newspaper page and feeds the image into the processing chain.

Component 1 handles image capture, preprocessing, and Sinhala headline recognition. The captured image passes through correction and preprocessing stages before a MAT Skeletonizer extracts the structural form of headline characters. A CNN Classification Model and Structural Recognition Model work in combination to identify headline text, with rule-based correction applied before the final headline characters are passed forward.

Component 2 receives both the Component 1 headline output and a parallel standard OCR text extraction stream. A clean Sinhala corpus is used to generate noisy-clean paired training data, which fine-tunes a transformer-based correction model. This model produces context-aware corrected Sinhala text that is then post-processed before being passed to Component 3 as structured input.

Component 3 receives corrected Sinhala text from Component 2 and matches it against a three-tier knowledge base verified permanent store, session cache, and web-scraped fallback through a two-stage retrieval pipeline. Retrieved content is stored in a local ChromaDB vector store and passed through a RAG pipeline connecting to Gemma 2B, while the Sinhala Complexity Classifier assigns a difficulty level that conditions the final output as either simplified or elaborated Sinhala text delivered to Component 4.

Component 4 receives the complexity-adapted Sinhala text from Component 3 and processes the user's voice input through a speech-to-text pipeline and intent detection model to identify what the user is asking for. It then applies personalization based on stored user preferences and interaction history before delivering the final response as natural Sinhala speech through a text-to-speech module.

3.2 System Architecture – Individual Component

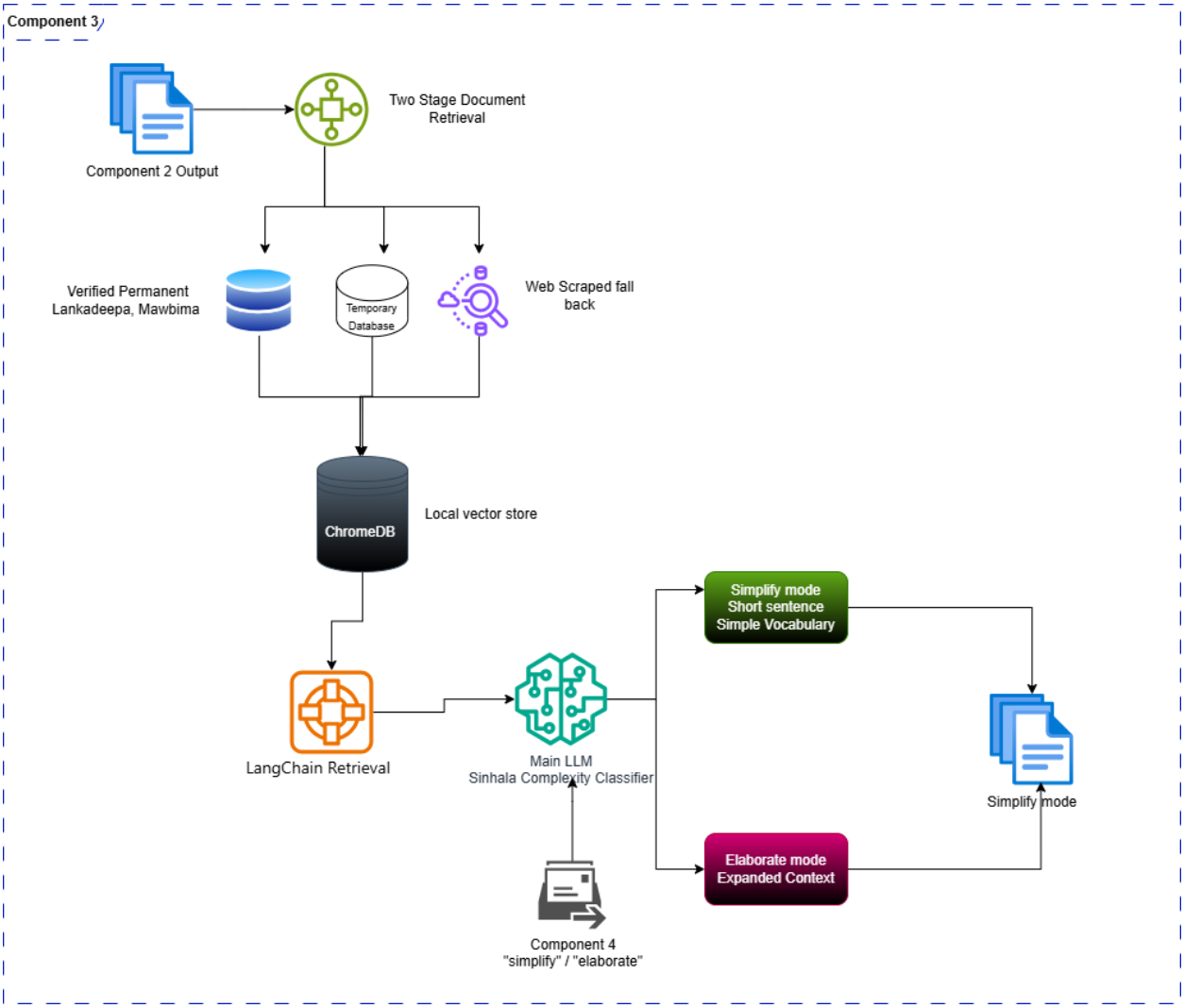


Figure 2. System Architecture

System Architecture

Figure 2 illustrates the system architecture of Component 3, which operates as a five-stage pipeline receiving corrected Sinhala text from Component 2 and delivering complexity-adapted content to Component 4.

The pipeline begins with Two-Stage Document Retrieval, where the incoming corrected Sinhala document is matched against the knowledge base through a two-step process headline embedding for coarse candidate selection followed by body-text cosine reranking for confirmation. The knowledge base itself operates across three tiers: a verified permanent store

populated from Lankadeepa and Mawbima, a temporary session cache, and a web-scraped fallback activated when no confident match is found in the permanent store.

All retrieved content flows into ChromaDB, a locally deployed vector store that serves as the single source of truth for the RAG pipeline. The RAG pipeline connects ChromaDB to the Main LLM Gemma 2B which generates contextually grounded Sinhala responses. Alongside generation, the Sinhala Complexity Classifier measures the difficulty level of the retrieved content and assigns a Level 1, 2, or 3 label.

When Component 4 sends a simplify signal, the system applies short-sentence and simple-vocabulary constraints to the output. When an elaborate signal is received, the system expands context through multi-article retrieval before generation. The final adapted Sinhala text is returned to Component 4 for text-to-speech delivery.

Category	Tool	Rationale
Programming Language	Python 3.11	Industry-standard NLP development environment with native support for all required libraries and Unicode Sinhala text handling
Sinhala NLP	sinling (UoM)	Only production-grade Sinhala tokenisation and morphological analysis library; essential for conjunct character processing and feature extraction
Embedding Model	LaBSE	Trained across 109 languages; preserves semantic similarity on OCR-degraded Sinhala input; 1.5GB RAM hardware feasible
Vector Database	ChromaDB	Persistent local vector store; offline operation; sub-100ms query latency; zero API cost; native Python integration
RAG Orchestration	LangChain	Native ChromaDB and Ollama integration; RetrievalQA chain handles context injection and prompt

		management in under 10 lines of code
Language Model	Gemma 2B via Ollama	Instruction-tuned; 1.5–2GB RAM requirement fits 8GB hardware budget; fully offline; single command deployment
Web Scraping	BeautifulSoup4	Lankadeepa and Mawbima serve static HTML; no browser automation needed; under 50MB memory overhead
Complexity Classifier	scikit-learn Logistic Regression	Generalises well on 120-sample dataset; linear feature relationships suit this model; learned weights are interpretable as linguistic findings
Data Handling	pandas + numpy	Standard data manipulation for feature extraction CSV and annotation dataset management

Table 2.Tools and Technology

3.2 Project Execution:

3.2.1 Data Collection Approach

The implementation of Component 3 follows a five-phase execution approach, each phase producing tangible outputs that feed directly into subsequent stages. The approach is designed to ensure that every technical claim made in the novelty statements is supported by a corresponding implementation artefact and a measurable evaluation result.

3.3.1 Data Collection Approach

News Article Corpus

Sinhala news articles will be systematically scraped from Lankadeepa and Mawbima using BeautifulSoup4 and Requests in Python 3.11, targeting their static HTML pages without browser automation. Each article will be stored as a structured JSON object capturing six fields article identifier, headline, section category, publication date, body text, and source URL. A minimum of 2,000 articles across five section categories will be collected, with hash-based deduplication applied to remove duplicate entries before knowledge base population.

Complexity Annotation Corpus

One hundred and fifty Sinhala paragraphs will be selected from the scraped corpus and Educational Publications Department textbooks, stratified equally across three register types of formal academic, standard news, and simplified community writing. Each paragraph will be presented to eight to ten native Sinhala speakers through a structured Google Forms instrument using a 1-to-3 difficulty rating scale, with a minimum of five independent ratings collected per paragraph. Mean ratings will be rounded to derive the final complexity label, and Cohen's Kappa will be calculated across all raters to formally establish inter-annotator agreement.

Dataset Purpose Mapping

The news article corpus forms the Tier 1 permanent store of the self-evolving knowledge base and constitutes the retrieval index against which all incoming scanned documents are matched. The complexity annotation corpus provides the labelled training data for the logistic regression complexity classifier the core artefact of Novelty 1. Both datasets are built entirely from scratch during this research and constitute original contributions independent of the system itself.

3.3.2 Data Preprocessing Approach

Text Normalization

All scraped Sinhala text will pass through Unicode NFC normalization using Python's Unicode data module, resolving encoding inconsistencies that arise from variation across different web sources. Zero-width joiner characters appearing incorrectly in scraped content will be removed programmatically, and residual HTML artefacts, excessive whitespace, and mixed-script characters will be stripped. The cleaned output will be stored as a clean body field alongside the original raw body field in each JSON object, preserving both versions for audit purposes.

Sinhala Tokenization

Sentence and word tokenization will be performed using the sinling library, the only production-grade Sinhala NLP toolkit available which handles Sinhala morphological complexity more reliably than generic multilingual tokenizers. This tokenization step is prerequisite to all four feature extraction operations and to the LaBSE headline embedding process. Sinling's tokenizer correctly handles conjunct character boundaries, which generic tokenizers consistently misidentify in Sinhala script.

Feature Extraction for Complexity Classifier

Four linguistic proxy features will be extracted from each preprocessed paragraph in the annotation corpus. Mean sentence length is computed as average word tokens per sentence using sinling; type-token ratio is computed as unique word types divided by total tokens measuring lexical diversity; conjunct character ratio is computed as the proportion of Sinhala HAL sign characters (U+0DCA)

to total characters serving as a formality proxy; and mean word length is computed as average Unicode code points per word token. These four values combined with the annotated complexity label form the five-column training matrix fed directly into scikit-learn for classifier development.

3.3.3 Model Selection and Development

3.3.3 Model Selection and Development

Selection Criteria

All model selection decisions across Component 3 are governed by three criteria applied consistently hardware feasibility on a Core i5 processor with 8 gigabytes of RAM, suitability for the specific task type, and availability of documented integration pathways. No model is selected on capability alone without confirming it fits within the available memory budget alongside all concurrently loaded components. This constraint eliminates several otherwise capable alternatives, including SinLLaMA and Llama 3.1 8B, before task suitability is even considered.

Embedding Model LaBSE

LaBSE is selected for headline and body-text embedding over monolingual English models on the grounds of its training across 109 languages and its demonstrated preservation of semantic similarity across linguistically distant language pairs essential for retrieval over OCR-degraded Sinhala input. It requires approximately 1.5 gigabytes of RAM, fitting comfortably within the available memory budget. No Sinhala-specific embedding model offers equivalent instruction-following capability within the hardware constraint.

Vector Database ChromaDB

ChromaDB is deployed as a locally persistent instance rather than a cloud-hosted service, driven by offline operation requirements, sub-100-millisecond query latency, zero API cost, and data privacy considerations for an assistive system handling personal reading behavior. Cloud alternatives such as Pinecone introduce latency, connectivity dependency, and per-query costs that are incompatible with the deployment context. ChromaDB's native Python integration with LangChain further reduces implementation complexity.

Complexity Classifier Logistic Regression

A multinomial logistic regression trained on 120 samples using scikit-learn is selected over Random Forest and transformer-based alternatives on three grounds superior generalization on small datasets, linear relationship between the four features and complexity level, and interpretability of learned weights as linguistic research findings. A Random Forest classifier will be trained on identical data as a comparison baseline to empirically justify this choice. The logistic regression model runs in milliseconds, adding negligible load to the already constrained hardware budget.

Language Model Gemma 2B

Gemma 2B served locally through Ollama is selected over SinLLaMA because SinLLaMA is a base model without instruction tuning it cannot reliably follow level-specific rewriting commands, which is the core requirement of the complexity adaptation module. SinLLaMA additionally requires a minimum of six gigabytes of RAM in 4-bit quantization, exceeding the available memory budget after other components are loaded. Gemma 2B requires only 1.5 to 2 gigabytes, is instruction-tuned, and deploys through a single Ollama command with native LangChain integration.

3.3.4 System Integration

Integration of Component 3 proceeds across four sequential stages, each validated before the next begins. The first stage batch-embeds all 2,000 scraped articles using LaBSE in groups of 64 and populates the tier1verified ChromaDB collection, verified through test queries on a ten-item spot-check set. The second stage connects the two-stage retrieval pipeline to the tier routing logic, directing queries with confidence below 0.60 to the BeautifulSoup web scraping fallback, with Tier 3 promotion verified through simulated repeated retrieval. The third stage loads the trained logistic regression classifier and connects it to the LangChain RetrievalQA chain, where the assigned complexity level conditions the prompt template before passing to Gemma 2B both simplification and elaboration modes are tested against ten manually prepared input-output pairs each. The fourth stage connects the complete pipeline to the mock Component 2 JSON interface, confirming the top-level component3 process function accepts the agreed input schema and returns adapted Sinhala text with a complexity level tag attached. Unit tests for each of the five internal stages are executed before end-to-end integration testing begins, ensuring any failure is isolated to its source stage.

3.3.5 Testing & Evaluation

Evaluation of Component 3 is built around five independently measurable results, each tied to a specific objective and a defined metric. The two-stage retrieval pipeline is benchmarked against a single-stage LaBSE baseline on thirty-five manually scanned OCR-noisy Sinhala newspaper articles, with Top 1 and Top 3 precision reported for both the improvement constituting Research Result 1. Knowledge base quality is measured as a retrieval precision curve recorded at five points across one hundred simulated sessions, demonstrating longitudinal improvement through the tier promotion mechanism as Research Result 2. The logistic regression complexity classifier and its Random Forest baseline are both evaluated on the thirty-sample held-out test set, with accuracy and per-class F1 reported in a unified comparison table as Research Result 3. A nine-participant native speaker user study spanning elderly, student, and expert demographic groups measures preference alignment rate between classifier-predicted complexity levels and participant-stated

preferences, constituting Research Results 4 and 5 alongside end-to-end latency measurement across twenty test sessions on target hardware against a fifteen-second threshold.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Primary Users: Visually impaired Sinhala-speaking adults who require hands-free access to printed newspapers and documents for daily information needs.

Secondary Users: Elderly individuals with age-related vision decline, individuals with limited literacy who comprehend spoken Sinhala but cannot read printed text.

Tertiary Stakeholders: Caregivers and family members who assist visually impaired people, special education teachers working with visually impaired students, and librarians managing accessible collections.

Technical Stakeholders: System maintainers responsible for updating the knowledge base, researchers in Sinhala NLP seeking to evaluate retrieval systems on real-world data.

5.2 Functional Requirements

1. Sinhala Document Reception The system shall receive a structured JSON object from Component 2 containing a corrected Sinhala headline, section label, and body sentences. All downstream processing depends on this input being correctly structured and Unicode-normalised.

2. Two-Stage Article Retrieval The system shall embed the incoming headline using LaBSE and retrieve the top five candidate articles from ChromaDB, followed by body-text cosine reranking to confirm the correct match. Retrieval confidence below 0.60 shall automatically trigger the web scraping fallback.

3. Web Scraping Fallback The system shall automatically scrape Lankadeepa or Mawbima using the article headline as a search query when no sufficient database match is found. The retrieved article shall be stored in Tier 3 with an initial reliability score of 0.3.

4. Tiered Knowledge Base Management The system shall maintain three ChromaDB collections verified permanent, session cache, and scraped fallback each governed by defined reliability scores and expiry conditions. Tier 3 articles reaching a reliability score of 0.8 through repeated successful retrieval shall be automatically promoted to Tier 1 without human intervention.

5. Complexity Classification The system shall extract four linguistic features from retrieved Sinhala text and classify it as Level 1, Level 2, or Level 3 using the trained logistic regression classifier. The assigned level shall condition all subsequent prompt construction for the generation pipeline.

6. RAG Response Generation The system shall inject retrieved article content into a LangChain RetrievalQA chain connected to Gemma 2B via Ollama and generate a contextually grounded Sinhala response. The generation pipeline shall operate fully offline with no external API dependency.

7. Simplification Mode Upon receiving a simplify signal from Component 4, the system shall apply Level 1 specific prompt constraints maximum ten-word sentences, simple vocabulary, active voice, and full preservation to rewrite the retrieved content. Output shall carry a Level 1 complexity tag.

8. Elaboration Mode Upon receiving an elaborate signal from Component 4, the system shall retrieve two additional semantically related articles from the same section category and inject all three as context for expanded generations. Output shall carry a Level 3 complexity tag.

9. Output Delivery The system shall return complexity-adapted Sinhala text with an attached complexity level tag to Component 4 within a target latency of fifteen seconds on standard consumer hardware. The output format shall conform to the agreed inter-component JSON schema.

5.3 Non-Functional Requirements

1. Performance and Latency The complete end-to-end pipeline shall deliver a complexly adapted Sinhala response within fifteen seconds per session on a Core i5 processor with 8 gigabytes of RAM. This will be achieved by loading all models at startup, batching ChromaDB queries, and limiting RAG context to three article chunks per generation call.

2. Memory Efficiency The total RAM consumption of all concurrently loaded components operating system, ChromaDB, LaBSE, LangChain, and Gemma 2B, shall not exceed 8 gigabytes on the target hardware. This will be achieved through model selection decisions that prioritise lightweight alternatives, specifically Gemma 2B over larger instruction-tuned models.

3. Offline Availability The system shall function without any internet connectivity during active user sessions, with all models, embeddings, and knowledge base collections stored and executed locally. This will be achieved by deploying ChromaDB as a persistent local instance and serving Gemma 2B through Ollama without external API calls.

4. Data Privacy No user reading data, voice queries, or scanned document content shall be transmitted to any external server or third-party service at any point during system operation. This

will be achieved through fully local deployment of all pipeline components with no telemetry or logging to remote endpoints.

5. Reliability The system shall handle low-confidence retrieval events gracefully by activating the web scraping fallback without interrupting the user session or producing an error state. This will be achieved through defined confidence threshold logic at 0.60 cosine similarity with automatic tier routing built into the retrieval function.

6. Maintainability Each of the five pipeline stages shall be implemented as an independently callable and independently testable function with defined input and output schemas. This will be achieved by writing unit tests for each stage before end-to-end integration testing begins.

7. Scalability the ChromaDB knowledge base shall support expansion beyond the initial 2,000-article Tier 1 corpus without requiring architectural changes to the retrieval or generation pipeline. This will be achieved through ChromaDB's native support for incremental document insertion and the metadata-based tier routing mechanism.

8. Accuracy The two-stage retrieval pipeline shall achieve measurably higher Top 1 precision than single-stage baseline retrieval on the thirty-five-article OCR-noisy ground-truth test set. This will be achieved through the headline-anchored coarse search followed by body-text cosine reranking that specifically targets OCR-induced vector shift.

9. Usability for Target Population The system shall produce output that is comprehensible to elderly, students, and expert user groups at their respective complexity levels, validated by a preference alignment rate reported from the nine-person native speaker user study. This will be achieved through the level-conditioned prompt templates in the simplification and elaboration modes.

10. Ethical Compliance The system shall collect informed consent from all user study participants before data collection and retain no personally identifiable information beyond the study period. This will be achieved through a structured consent process administered before each user study session with all responses anonymized at the point of recording.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target users are visually impaired and elderly Sinhala-speaking individuals in Sri Lanka who rely on printed newspapers as their main source of daily information. This includes people with age-related vision loss, congenital visual impairment, and acquired vision disabilities across both urban and rural communities. The secondary market comprises Sinhala NLP researchers, assistive technology developers, educational institutions, and government bodies such

as the Ministry of Education and the Sri Lanka National Council for the Blind who require complexity-adaptive Sinhala content delivery infrastructure for broader accessibility initiatives.

Value Proposition

The system delivers three capabilities that no existing commercial or open-source product currently provides simultaneously for Sinhala OCR-robust newspaper article retrieval, self-evolving knowledge base management, and reader-adapted content delivery at three complexity levels. The Sinhala Text Complexity Framework produced by this research is the first of its kind, offering a reusable, empirically validated tool for any Sinhala NLP application requiring reading difficulty measurement. The hardware cost of deployment falls under $50 compared to $1,800 or more for commercial alternatives making the system accessible to the demographic most in need of it.

Revenue Model

The core system will be released as an open-source module under an MIT or Apache 2.0 license, enabling free adoption by the Sinhala NLP research community, assistive technology developers, and educational publishers. This open release ensures the widest possible benefit to visually impaired end users without financial barriers. Commercial revenue may be generated through a separate channel providing detailed technical documentation, integration support, and consulting services to organizations wishing to embed the component into existing products or institutional workflows at a negotiated fee.

Cost Estimation

The primary hardware cost is a glasses-mounted USB camera capable of capturing sufficient image quality for downstream OCR processing estimated at $25 to $35 for a suitable autofocus compact webcam. All software components ChromaDB, LaBSE, LangChain, Gemma 2B via Ollama, BeautifulSoup, sinling, and scikit-learn are open source and carry no licensing cost. The processing hardware requirement is a standard consumer laptop, which the target user is assumed to already possess. Total additional hardware expenditure is therefore under $50 per deployment.

Pricing Strategy

The component itself carries no price. The open-source release strategy prioritizes adoption and community-driven improvement over direct monetization. Detailed architecture documentation, annotated datasets including the 150-paragraph Sinhala complexity corpus and integration guides

may be offered at a cost to commercial entities seeking to embed the system into proprietary products, alongside a commercial use agreement where applicable.

Competitive Advantage

- Only known system providing complexity-adaptive Sinhala document retrieval validated through a native speaker user study.
- Self-evolving knowledge base eliminates the manual curation burden that limits every existing Sinhala NLP system.
- Fully offline deployment protects user privacy and removes internet dependency critical for rural and low-connectivity users.
- Modular pipeline design allows individual components retrieval, complexity classifier, RAG generation to be adopted independently by other systems.
- The annotated Sinhala complexity corpus is a reusable research asset that creates academic and commercial value beyond the system itself.
- Total deployment cost under $50 positions the system as the only financially accessible Sinhala assistive reading solution for the target demographic.

Intellectual Property Considerations

No patent protection is sought. The novel contributes the Sinhala Text Complexity Framework, the headline-anchored two-stage retrieval pipeline, and the tiered self-evolving knowledge base will be documented in academic publications with attribution requirements enforced through the chosen open-source license. The 150-paragraph annotated complexity corpus will be released under a Creative Commons license to encourage extension and reuse by the Sinhala NLP research community.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost	Justification
Hardware (External)	Camera, Microphone	LKR 7,000 upwards	These are essential to build the system and even if we use pre-shot images, the better camera directly improves the project results.
Software	All tools open-source (Python, OpenCV, scikit-image)	-None	All primary tools are freely available
Data	Extracted text from the images of the Sinhala document.	-Depends, as documents to capture images can be bought or borrowed.	Ground truth dataset creation
Man-Hour	Student research time	-Not applicable	Already accounted for in the project
Hardware (Internal)	Computer	-Not applicable	Currently available laptops will be mostly available to the software.

Table 3. Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

Initial Phase	Planning	Data Preparation & Input	Retrieval & RAG Pipeline	Testing & Evaluation
Need Identification: OCR noise & context gaps	**Retrieval Workflow**: Designing the multi-source pipeline	PDF Collection Lankadeepa & Dinamina	**Text Chunking**: Recursive character splitting	**Retrieval Accuracy**: Testing similarity thresholds (0.85)
Feasibility Study: Assessing SinLLaMA & ChromaDB	**Tech Selection**: SinLLaMA-7B and PyMuPDF	Dataset Labeling	**Sinhala Embeddings**: SinLLaMA-7B semantic vectoring	**Context Completeness**: Target 99.8% recovery vs 60% OCR
Data Acquisition: Defining official print replica sources	**Metadata Design**: Date, source, and URL mapping		**Vector Indexing**: ChromaDB persistent store setup	**Performance**: Measuring <50ms retrieval latency

			RAG Generation	**Response Quality**: Testing simple vs. elaborate levels
			Dynamic Adaptation: Integration of user profiles	

Table 4. Work Break Down Structure

9. GANTT CHART

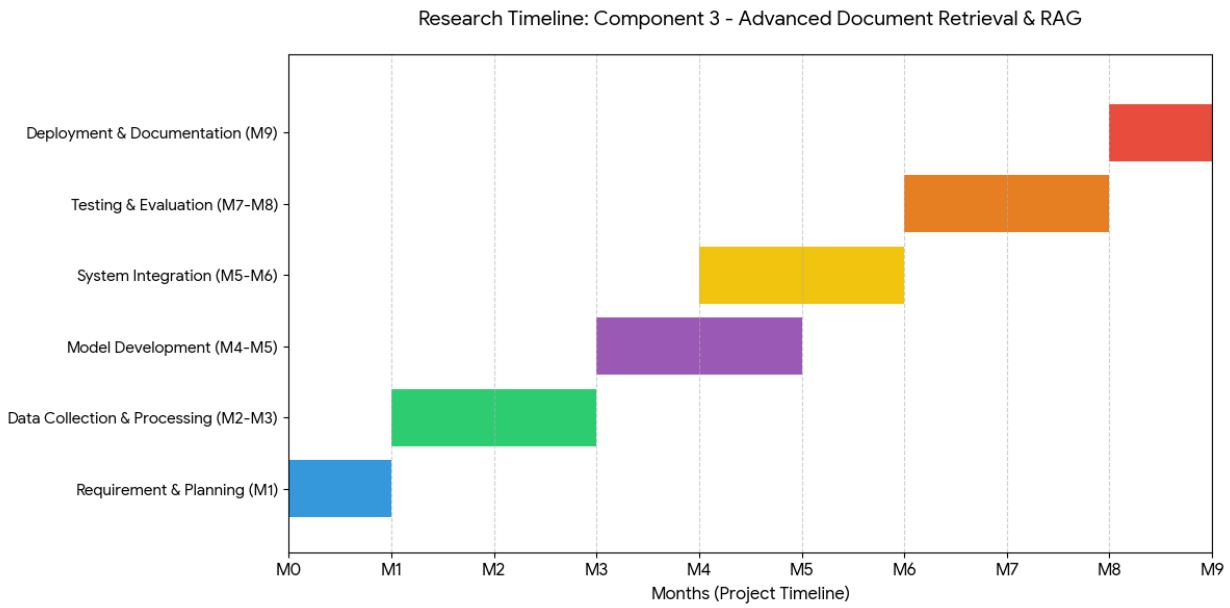


Figure 3.Gantt Chart

M0 – January
M1 – February
M2 – March
M3 – April
M4 – May
M5 - June
M6 - July
M7 - August
M8 - September
M9 – October

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11. APPENDICIES

11. 1 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Web scraping blocked by Lankadeepa or Mawbima rate limiting	Medium	High	Implement request delays between scrapes; maintain pre-scraped offline backup corpus as contingency
Gemma 2B producing inconsistent Sinhala output for level-conditioned prompts	Medium	High	Iteratively refined Sinhala prompt templates; test across twenty evaluation cases before integration
ChromaDB memory consumption exceeding 8GB hardware budget when loaded alongside LaBSE and Gemma 2B	Medium	High	Batch-load models sequentially; reduce ChromaDB collection size if necessary; profile memory at each integration stage
Component 2 JSON schema delivered late or incompatible with Component 3 input interface	High	Medium	Build and develop against mock Component 2 JSON from Week 1; agree schema with group in Month 1
LaBSE retrieval precision insufficient on OCR-noisy Sinhala headlines	Low	Medium	Fall back to exact headline substring matching as secondary retrieval signal; tune confidence threshold

Table 5.Risk Analysis

11.2 Ethical Document

This research involves human participants in the complexity annotation task and the user study validation phase. Participants in the annotation task will be asked to rate the difficulty of written Sinhala paragraphs through a Google Forms instrument. Participants in the user study will interact with the system and indicate content complexity preferences across three adapted text samples.

Informed consent will be obtained from all participants prior to any data collection activity. Participants will be clearly informed of the research purpose, their right to withdraw at any point without consequence, and how their responses will be used. No personally identifiable information will be retained beyond the study period. All collected responses will be anonymized at the point of recording and stored securely on a password-protected local device accessible only to the researcher.

User study involves no clinical observation, medical intervention, or collection of sensitive personal data. Participation is entirely voluntary and involves no physical risk. The research is conducted under SLIIT Faculty of Computing undergraduate research ethics guidelines and does not require formal ethics board clearance.

Data sourced through web scraping Sinhala news articles from Lankadeepa and Mawbima is publicly available content accessed through standard browser requests. No paywalled, private, or personally identifiable content will be collected. The annotated complexity corpus produced during this research will be released under a Creative Commons license for academic use, with no participant-linked information included in the release.